Q1.

A long time ago sulphuric acid was made by heating a substance called **blue vitriol**.
The equations below show how sulphuric acid is produced by this method.

blue vitriol ⟶ copper oxide + sulphur trioxide + water

sulphur trioxide + water ⟶ sulphuric acid

(a) Name **three** elements contained in blue vitriol.

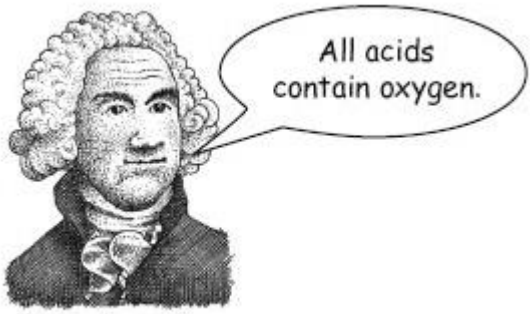
1. .....

2. .....

3. .....

3 marks

(b) (i) Anton Lavoisier was a scientist. He made acids by dissolving oxides like sulphur oxide and nitric oxide in water. They formed two acids; sulphuric acid and nitric acid. From this, he concluded:



Anton Lavoisier

The formulas for these two acids are H_2SO_4 and HNO_3.
How do these formulas support Lavoisier's conclusion about acids?

.....

.....

1 mark

(ii) Some time after Lavoisier's death, hydrochloric acid was identified.
The formula for hydrochloric acid is HCl.

Explain why scientists no longer supported Lavoisier's conclusion about acids.

.....

.....

1 mark

(c) Scientists now agree that **all** acids contain hydrogen.
Look at the two word equations below.

zinc + sulphuric acid ⟶ zinc sulphate + hydrogen

magnesium + nitric acid ⟶ magnesium nitrate + hydrogen

(i) Explain how these equations support the suggestion that acids contain hydrogen.

.....

.....

1 mark

(ii) Complete the equation below for the reaction between iron and hydrochloric acid.

iron + hydrochloric acid ⟶ ..... + ...

1 mark

maximum 7 marks

Q1.

(a) any **three** from

- copper
- oxygen
- sulphur
- hydrogen

accept chemical symbols
'Cu'
'O'
'S'
'H'
*do **not** accept 'O₂' **or** H₂'*

3 (L6)

(b) (i)
- they both contain oxygen

accept 'they both have O in them'
*do **not** accept 'they have O₂ in them'*

1 (L6)

(ii) any **one** from

- hydrochloric acid does not contain oxygen

accept 'it does not contain oxygen'
'he was wrong' is insufficient

- hydrochloric acid is not formed from oxygen

accept 'not all acids contain oxygen'
accept 'no oxygen'

1 (L6)

(c) (i) any **one** from

- they both produce hydrogen (when they react with metals)
- different metals produce the same gas

accept 'there is hydrogen left at the end'
'they contain hydrogen' is insufficient
'hydrogen is in the equation' is insufficient

1 (L6)

(ii)
- iron chloride
  hydrogen

*accept 'FeCl₂' **or** 'FeCl₃' **or** 'Cl₂Fe'*
accept 'H₂'
***both** answers are required for the mark*
answers may be in either order
*do **not** accept 'iron + chloride'*

1 (L6)

[7]